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IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

ARTI 308 — MACHINE LEARNING FINAL PROJECT

# AI-BASED CREDIT CARD FRAUD DETECTION SYSTEM

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# MOTIVATION & BACKGROUND

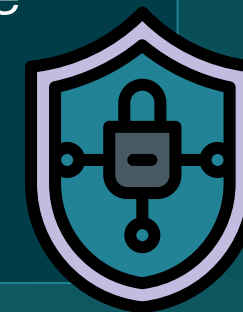
## Rising Fraud Threat

Threat Rapid expansion of digital payments has significantly increased credit card fraud worldwide, causing major financial losses



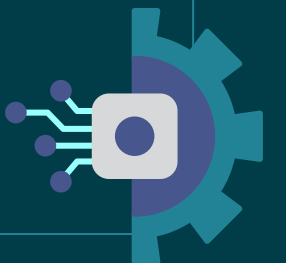
## Inadequate Defenses

Traditional rule-based systems fail to detect new and evolving fraud techniques in real-time environments



## ML as a Solution

Machine learning can analyze massive transaction datasets, identify hidden patterns, and adapt to new fraud behaviors





# PROJECT OBJECTIVES

- 01 Develop and compare three ML classification models for fraud detection
- 02 Address severe class imbalance using SMOTE oversampling technique
- 03 Evaluate models using Accuracy, Precision, Recall, F1-Score, and AUROC
- 04 Identify the most effective and balanced fraud detection model
- 05 Investigate the impact of feature selection on classification performance



# CLASS IMBALANCE PROBLEM

284,807

Total Transactions

492

Fraudulent Transactions

0.17%

Fraud Percentage



## WHY CLASS IMBALANCE IS A PROBLEM

- Most ML models tend to favor the majority class, leading to poor fraud detection.
- Accuracy alone is misleading — a model predicting everything as legitimate gets 99.83%!
- Need metrics: Precision, Recall, F1, and AUROC
- Solution: **SMOTE** generates synthetic fraud samples to balance the training data.

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# DATASET CHARACTERISTICS

284,807

Total Transactions

492

Fraud Cases

30

Features

0.17%

Fraud Rate

## DATASET OVERVIEW

Source: Kaggle – European cardholders, September 2013 (2-day period)

Features: V1–V28 (PCA-anonymized) + Time + Amount + Class (target)

Target: Class 0 = Legitimate | Class 1 = Fraudulent

Privacy: Original transaction details transformed via PCA to protect cardholder confidentiality



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# MACHINE LEARNING WORKFLOW

1

Data  
Loading

Kaggle Credit Card  
Fraud Dataset

2

Data  
Preprocessing

Data Cleaning &  
EDA

3

Feature  
Scaling

StandardScaler  
Applied

4

SMOTE

Class Imbalance  
Handling

5

Model  
Training

LR, DT, RF Models

6

Hyperparameter  
Tuning

GridSearchCV

7

Model  
Evaluation

Acc, Prec,  
Rec, F1, AUROC

**Each step was applied within a reproducible pipeline with fixed random seeds and stratified train-test splits.**



# WHY SMOTE?

## WHAT IS SMOTE?

SMOTE generates synthetic fraud samples based on existing minority-class transactions, helping balance the dataset without simple duplication.

## KEY RULES IN THIS STUDY:

- Applied to TRAINING data only
- Prevents data leakage to test set
- Integrated within ML pipeline

## BEFORE SMOTE

**Legitimate Transactions:**  
**284,315**

**Fraud Transactions:**  
**492 (0.17%)**



## AFTER SMOTE

**Balanced Training Data**

- ↑ Improved Fraud Detection
- ↑ Higher Recall Performance



# ALGORITHM SELECTION

## Logistic Regression

BASELINE

### STRENGTHS

- + Simple & interpretable
- + Computationally efficient
- + Good baseline benchmark

### LIMITATIONS

- Limited non-linear capability
- Poor precision on imbalanced data

## Decision Tree

COMPARISON MODEL

### STRENGTHS

- + Handles non-linear patterns
- + Interpretable rules
- + No feature scaling needed

### LIMITATIONS

- Prone to overfitting
- Unstable on noisy data

## Random Forest

PRIMARY CANDIDATE



### STRENGTHS

- + Ensemble reduces overfitting
- + Strong with imbalanced data
- + Highest overall performance

### LIMITATIONS

- Higher computational cost
- Less interpretable

★ BEST CHOICE: Random Forest achieved the best overall balance between Precision (92.4%), Recall (76.8%), F1-Score (83.9%), and AUROC (0.9519).



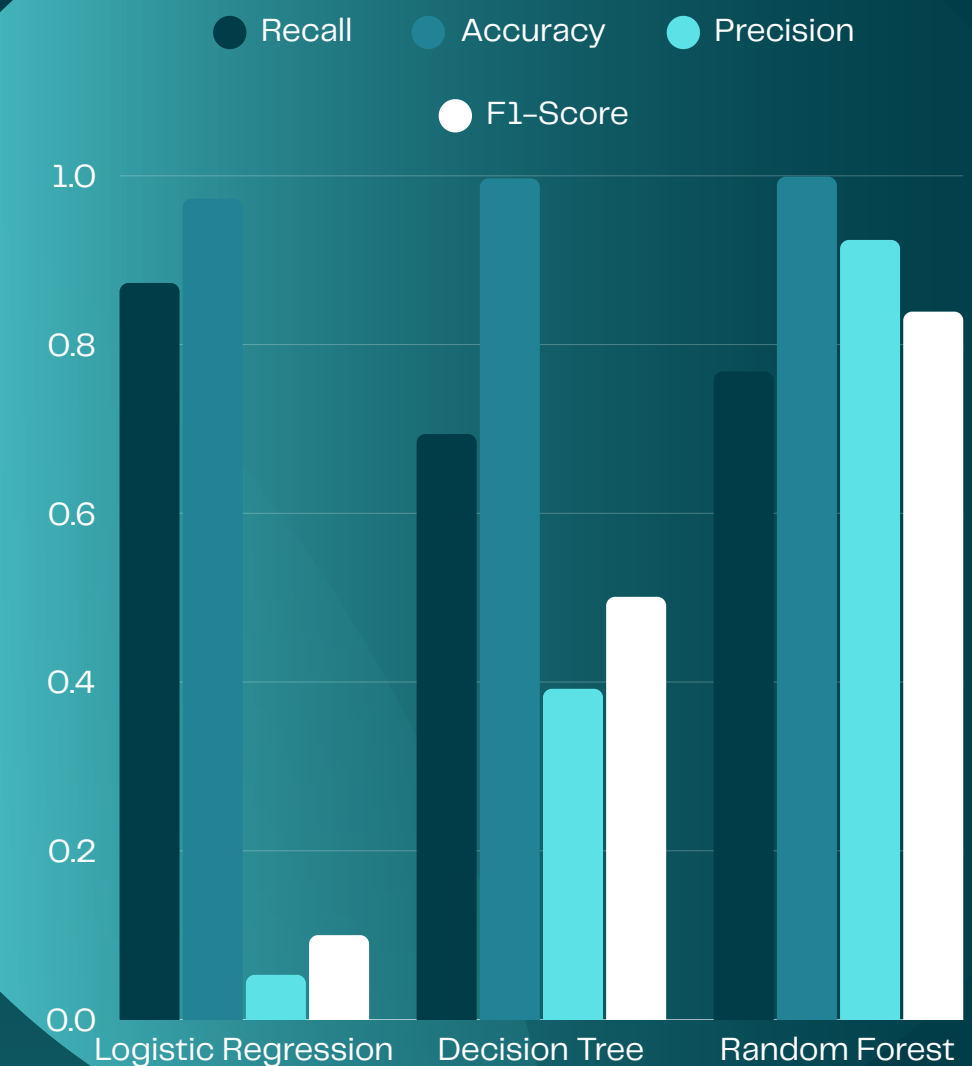


# PERFORMANCE METRICS COMPARISON

- Logistic Regression: Highest Recall (87.3%) but very low Precision (5.3%)
- Decision Tree: Improved precision but moderate overall performance
- Random Forest: Best overall balance between Precision (92.4%) and Recall (76.8%)

| Model               | Accuracy | Precision | Recall | F1-Score |
|---------------------|----------|-----------|--------|----------|
| Logistic Regression | 0.973    | 0.053     | 0.873  | 0.1      |
| Decision Tree       | 0.997    | 0.392     | 0.694  | 0.501    |
| Random Forest       | 0.999    | 0.924     | 0.768  | 0.839    |

F1-Score is especially useful for evaluating imbalanced datasets.





# CRITICAL DISCUSSION

## HIGH ACCURACY $\neq$ HIGH RECALL

Accuracy of 99.95% is impressive but Recall of 76.84% means approximately 23% of fraud cases remain undetected, which represents a critical risk

## PRECISION VS RECALL TRADE-OFF

Logistic Regression achieves 87.4% recall but only 5.3% precision — flooding analysts with false alarms. Random Forest balances both at 92.4% / 76.8%.

## SMOTE EFFECTIVENESS

Without SMOTE, models would classify everything as legitimate and achieve 99.8% accuracy trivially. SMOTE forces models to learn fraud patterns.

## REAL-WORLD LIMITATIONS

22 missed fraudulent transactions in the test set represent potential real-world financial losses. Future work should focus on improving recall while maintaining high precision.

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# CONFUSION MATRIX ANALYSIS

|               | Predicted: Legitimate                      | Predicted: Fraudulent                  |
|---------------|--------------------------------------------|----------------------------------------|
| Actual: Legit | <div>56,645</div> <div>True Negative</div> | <div>6</div> <div>False Positive</div> |
| Actual: Fraud | <div>22</div> <div>False Negative</div>    | <div>73</div> <div>True Positive</div> |

**Precision:** 92.41%

Only 6 false alarms out of 79 predicted frauds

**Recall:** 76.84%

22 of 95 fraud cases were missed

**False Positive Rate:** 0.011%

Virtually no legitimate transactions flagged

**Miss Rate:** 23.2%

Requires improvement for real-world deployment



# AUROC ANALYSIS

**0.9519**

**AUROC**  
Area Under ROC curve

## PERFORMANCE SCALE

**0.95–1.00**

**Outstanding**

**0.90–0.95**

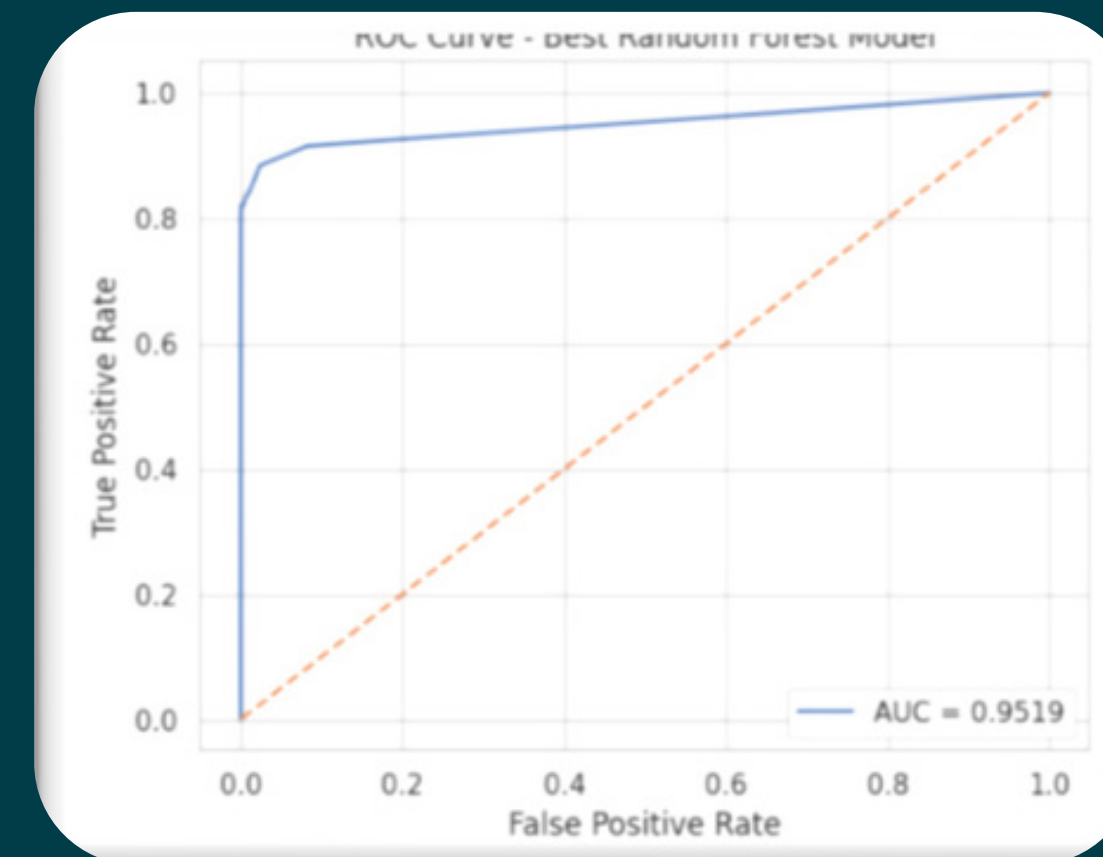
**Excellent**

**0.80–0.90**

**Good**

**< 0.80**

**Fair / Poor**



## What is AUROC?

- Measures the model's ability to distinguish between fraudulent and legitimate transactions.
- Values range from 0.5 (random guessing) to 1.0 (perfect discrimination)

## Result

- Random Forest achieved an AUROC of 0.9519.

## Interpretation

- Indicates excellent discrimination capability.
- The model can correctly rank fraudulent transactions higher than legitimate ones in most cases.
- Confirms that Random Forest is effective even under severe class imbalance.

## Key Insight

- High AUROC demonstrates strong overall classification performance beyond accuracy alone.



# CONCLUSION & FUTURE WORK

Best model random forest

**99.95%**  
Accuracy

**92.41%**  
Precision

**76.84%**  
Recall

**83.91%**  
F1-Score

**0.9519**  
AUROC

## CONCLUSION

- Three machine learning models were evaluated:
- Logistic Regression
- Decision Tree
- Random Forest
- SMOTE successfully addressed the severe class imbalance problem.
- Random Forest achieved the best overall performance.
- Random Forest provided the best balance between fraud detection and false alarms.

## FUTURE WORK

- Explore deep learning approaches.
- Test advanced ensemble methods such as XGBoost and LightGBM.
- Apply real-time fraud detection systems.
- Use larger and more recent datasets.
- Improve recall while reducing false positives.



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THANK YOU

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